Operating Instructions

for

Flowmonitor

Model: DF-WM

KOBOLD

1. Note

Please read and take note of these operating instructions before unpacking and commissioning. The instruments may only be used, maintained and installed by personnel familiar with the Operating Instructions and the applicable Health and Safety Requirements.

2. Contents

3. Declaration of Conformance

We, KOBOLD-Messring GmbH, Hofheim-Ts, Germany, declare under our sole responsibility that the product:

Flowmonitor Model DF-WM

to which this declaration relates is in conformity with the standards noted below: **EN 50082-2**

Electromagnetic compatibility (EMC) - Generic Immunity Standard

EN 61010

Safety requirements for electr. equipment for measurement, control and laboratory use.

following the provision of European Directives:

89/336/EWG and 73/23/EWG

Signed H.-D. Nemyczuk
H.D. Nemyczuk

Date: 14.04.1997

Manufactured and sold by:

Kobold Messring GmbH .... Tel.: 06192-2990
Nordring 22-24 .... Fax: 06192-23398
D-65719 Hofheim

4. Specific Application

The model DF-WM is installed to monitor and to measure discrete the flow of liquids.

The instrument provides the following facilities:

- **Discrete flow measurement**
  By means of a button and a setpoint selector the actual flow through put can be called up (see flow throughput measuring).
- **Limit value relays**
  The instrument is provided with limit value relays for monitoring flow throughput.

It is suitable for low viscosity fluids which have no effects on the instrument materials used. If using higher viscosity media large deviations willl occur from the flow range as given in the catalogue. Long threads can lead to the seizure of the rotor.
Likewise, ferritic particles can build up on the rotating vane and lead to faulty operation or destruction of the rotor. In cases of doubt, please contact the supplier.

Material Combinations

	Standard Version						High Pressure Version	
Material-Combination	I	II	II 8*	III	IV*	V*	VI*	VII*
Connection Type	Threaded	Threaded	Threaded	Threaded	Threaded Flanged	Threaded	Threaded	Threaded Flanged
Housing	Trogamid	Polysulfon	Polypropylen	brass, nickel plated	st.steel	PTFE	brass, nickel plated	st. steel
Housing Cover	Trogamid	Polysulfon	Polypropylen	Polysulfon	Polysulfon	PTFE	brass, nickel plated	st. steel
Connection	brass nickel plated	st. steel	Polypropylen	brass, nickel plated	st. steel	PTFE	brass nickel plated	st. steel
Securing Pins	brass	brass	brass	brass	–	–	–	–
O-Ring	Buna-N	Viton	Viton	Buna-N	Viton	–	Buna-N	Viton
Rotor	POM	PTFE	PTFE	POM	PTFE	PTFE	POM	PTFE
Axes[1]	st.steel	st.steel	ceramic	st.steel	st.steel	ceramic	st.steel	st.steel
Axes Bearings[1]	PTFE	PTFE	PTFE	PTFE	PTFE	PTFE	PTFE	PTFE
Orifice	PTFE[2]	PTFE[3]	PTFE[3]	PTFE[3]	PTFE[3]	PTFE[3]	PTFE[3]	PTFE[3]
max. Pressure (bar)	10	10	6	16	16	6	100	100 (Flange 40)
max. Temperature	80°C	80°C	80°C	80°C	80°C	80°C	80°C	80°C

1 Special with saphire axis and bearings

2 for DF 0,5 st.steel model

3 for DF 0,5 titanium model

* connection not rotable

5. Operating Principles

A plastic rotating vane rotates on an axle when a flow throughput occurs. A ring shaped magnet hermetically sealed in the rotating vane transmits this rotary motion to a Hall sensor mounted outside on the instrument housing. The electronics mounted on the housing converts the frequency signal and activates a limit switch.

6. Instrument Inspection

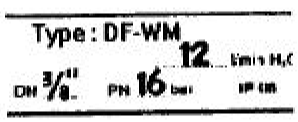
The instruments are inspected before dispatch and sent out in perfect condition. Should damage to the instrument be visible, we recommend close inspection of the delivery package. In cases of damage, please immediately inform the forwarder as he is liable for any damage in transit.

Scope of Supply

- All parts belonging to the standard scope of supply are attached to the instrument.
- For instruments with socket connection the coupling plug is also supplied.

7. Mechanical Connection

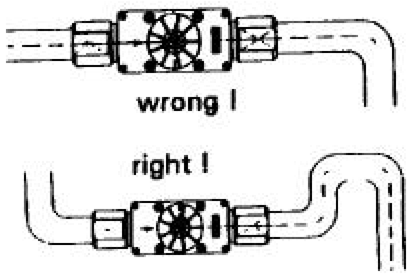
Before installation:



- Please ascertain whether the actual flow throughput matches the flow range of the instrument. The flow range may be obtained from the label.
-

Warning! If the measuring range is exceeded by more than 20%, bearing damage may occur.

- Please ascertain whether the allowable maximum operating pressure and operating temperature of the instruments will not be exceeded.
- Make sure that the electrical supply to the instrument conforms with the equipment operating data.
- Remove all transport packing and ascertain that no packing material is left in the instrument.

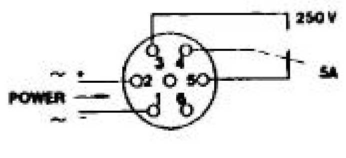


- The instrument may be installed in any position. However, the flow must always take place in the direction of the arrow, while the front face of the instrument must always be in the vertical plane.
- It must be ensured that the instrument housing is continuously filled with the flow medium, especially for flows from top to bottom. No straight pipe lengths are necessary at inlet and outlet connections.
- Sealing of the connection threads should be carried out with Teflon tape or similar.
- During installation of the instrument, it must be checked that no stress is applied to the connections. We recommend that the inlet and outlet pipes are mechanically fixed approximately 50 mm from each instrument connection.
- When using Material Combination IIB,IV,V,VI and VII the instrument connections may not be rotated.
- Check that the connection thread to pipe is fully sealed.

Warning ! The threaded connections of the instrument must be tightened with a suitable sized open ended spanner. Otherwise, the housing may be stressed which could lead to breakage of the equipment.

8. Electrical Connection

Warning! Make sure that the supply voltage to your instrument conforms with the value given on the equipment label.



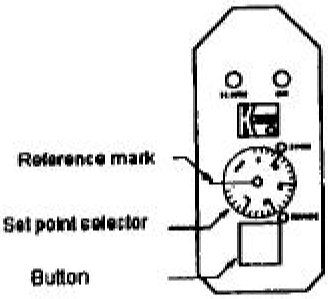
Cable connection:
No.1 : Power supply (-)
No.2 : Power supply (+)
No.3:
No.4:
No.5:

- Ensure that the power is disconnected during connection of the cable.
- For equipment with plug connection solder the ends of your supply cables in accordance with the connection plan supplied with the coupling plug.
- For equipment with cable connection simply connect the instrument cable to your supply cable.
- Supply cable cross-section: 0.75 mm².

Warning! Incorrect wiring of the connection in the coupling plug or incorrect wiring of the connection cable can lead to the electronics being destroyed.

- Plug the coupling plug in the socket provided on the equipment (in the case of instruments with plug connections).
- After connection of your external equipment at the connection points the instrument is ready for operation.

9. Electrical Commissioning of the Instruments



The instrument is delivered ready to operate. The electronics are matched and calibrated against the signal transmitter. The calibration screws ('ZERO' and 'RANGE') next to the rotating scale adjustment button must not be adjusted by the customer. If they are adjusted, a re-calibration will be required involving new calculations. Should the eletronics be opened any warranty will become invalid.
As soon as the external power supply to the instrument is switched on, a green LED will indicate that the unit is ready for operation. The red 'ALARM' light diode will flash.

Warning! A red, flashing LED indicates that the limit value relay is in alarm condition (see connection diagram).

Limit value

The Type DF-WM instrument has a limit value relay for monitoring the flow throughput. The switch condition of the relay is indicated by the flashing, red LED.The contact is designed as a minimum contact; i. e. for flow less than the selected flow value the limit value relay will indicate an alarm condition (flashing, red LED).

Warning! The limit value relay will indicate an alarm condition if the supply cable is broken or if power supply is interrupted.

Adjustment of limit value

- Loosen the four screws on the front plate of the electronics and remove the transparent cover.
- The limit value may be adjusted by means of the set point selector.
- The required limit value may be set by turning the set point selector so that the value aligns with the reference mark to the left of the scale on the front plate of the electronics.

Call-up of actual flowrate

To measure the actual flowrate the red button must be depressed. The limit value relay is then bridged and made inoperative. By rotating the set point selector at the same time from the lowest to the highest value (until the red light flashes) the flow rate may be determined from the scale or the set point selector. After reading off the flow rate the set point selector may be repositioned to the desired limit value and the red button released.

Warning! After adjustment of the switch or call up of the flow throughput the frontplate must be screwed down tight onto the electronics housing. Ensure correct positioning of the gasket.

10. Commisioning of the Instrument

To avoid pressure surges, the flow medium should be slowly introduced into the instrument.

Warning! Pressure surges from solenoid valves, ball valves or similar may lead to breakage of the instrument (water hammer). In the operating condition it must be checked that the instrument housing is continuously filled with the flow medium.

Warning! Large air bubbles in the instrument housing can lead to measuring errors or destruction of the bearings.

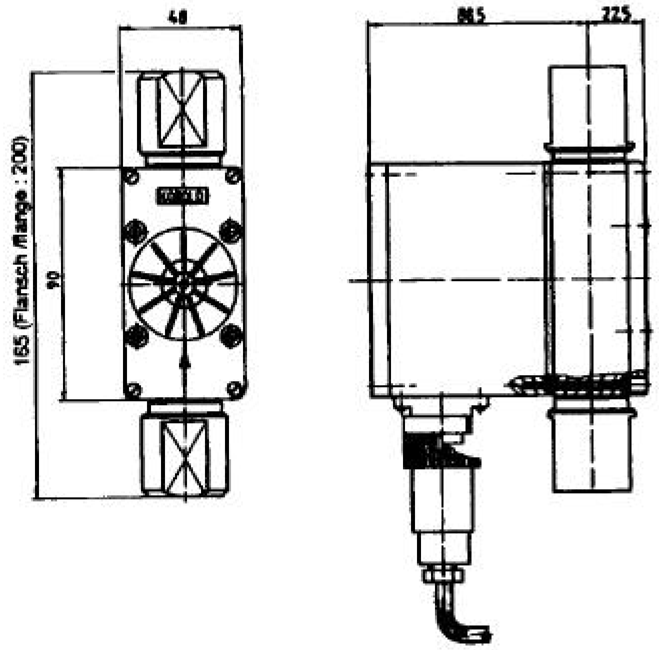
11. Technical Details

Power requirement:	3,5 W max.
Supply voltage:	optional (see label): 24 VDC, ±20% 24 VAC, 110 VAC, 230 VAC 50-60 Hz ±20 %
Max. contact load:	250 V max. / 5 A max.
Internal resistance:	< 100 mΩ
Protection:	IP 65
Switch accuracy:	±2,5% full scale
Ambient temperature:	-25° C bis +80° C
Max. pressure:	see material combination table (page 3)
Max. medium temperature:	see material combination table (page 3)

12. Maintenance

For measured media without contamination, the DF-WM instrument is almost maintenance-free. As the rotating vane contains magnets, any ferritic particles present in the medium may lead to problems. In order to avoid such problems we recommend the installation of a magnet filter e.g.: mpdel MF-R. Should cleaning of the instrument become necessary, the housing cover may easily be removed to provide access to the internals. After cleaning, the instrument may easily be reassembled. Any work on the electronics may only be undertaken by the supplier, otherwise the warranty will become invalid.

13. Dimensions



14. Recommended Spare Parts

1.1) Rotating vane PTFE
1.2.) Rotating vane POM
1.3) Rotating vane PTFE with saphire bearings
2.1.) Stainless Steel axle / PTFE bearings
2.2.) Ceramic axle/ PTFE bearings
2.3) Saphire axle with saphire bearings (only for rotating vane 1.3)
3.1.) Cover for instrument housing, Trogamide, including seal
3.2.) Cover for instrument housing, Polysulphone, including seal
4.1.) Transparent cover for electronics housing
5.1.) Set of Perbunan 0-rings
5.2.) Set of Viton 0-rings

For spare part ordering we need the serial no. of the instrument.